

Milestone 1: Project Overview

The **Expense Tracker** is a mobile application designed to help users take control of their personal finances by providing an intuitive platform for recording, categorizing, and analyzing daily expenses. In today's fast-paced world, managing finances effectively requires more than just awareness; it demands accessible tools that make tracking spending effortless and insightful. This app aims to bridge that gap by offering users a streamlined way to monitor their financial habits, identify spending patterns, and make informed decisions about their budgets. The primary goal is to create a user-friendly experience that encourages consistent expense logging while providing meaningful visual feedback on spending behavior across different categories.

The application will be built using React Native and Expo, ensuring cross-platform compatibility for both iOS and Android devices. The user interface will feature four main screens, each serving a distinct purpose in the expense management workflow. The Home Screen will serve as the central dashboard, displaying the user's total spending alongside a scrollable list of recent transactions, giving users immediate insight into their financial activity. The Add Expense Screen will provide a comprehensive form where users can input transaction details including the expense amount, category selection, optional description, and date, with built-in validation to ensure data accuracy. The Category Screen will offer analytical insights by presenting total spending broken down by category, potentially enhanced with visual charts to help users quickly identify their largest spending areas. Finally, the Settings Screen will allow users to personalize their experience through theme customization, including a dark mode toggle, and provide utility functions such as clearing all stored data.

To create a polished and functional application, the project will leverage a variety of React Native components and modern development practices. Navigation between screens will be handled using React Navigation with a Tab Navigator, providing users with quick access to

all major features. The app will incorporate TextInput components with validation logic to ensure expense data is entered correctly, while FlatList components will efficiently render potentially long lists of transactions. Performance optimization will be achieved through React hooks such as useEffect for managing side effects like data persistence and useMemo for calculating expensive operations like category totals without unnecessary re-renders. Visual appeal will be enhanced with LinearGradient backgrounds and icons from the Expo vector icons library, creating a modern and engaging interface.

State management and data persistence are critical components of this application's architecture. The Context API will be implemented to manage app-wide state, including theme preferences and calculated spending totals, ensuring consistency across all screens without prop drilling. AsyncStorage will handle local data persistence, allowing users to close and reopen the app without losing their expense history. Alert dialogs will be used for important user interactions, such as confirming deletion of expenses or clearing all data, preventing accidental data loss. The application will also feature a custom font to establish a unique visual identity, a splash screen for professional app loading, and a centralized theme constants file to maintain consistent styling throughout the codebase. Responsive design principles will ensure the app adapts seamlessly to different screen sizes and orientations, providing an optimal user experience regardless of device. Through this combination of thoughtful design, robust functionality, and technical excellence, the Expense Tracker will deliver a comprehensive solution for personal financial management.

Expense Tracker - App Mockup

